ActInf GuestStream 067.1 ~ Andrés Corrada "The logic of NTQR evaluation

GuestStream | Andrés Corrada | 1:51:15

Hello and welcome. It's December 7th, 2023. We're here in Active Inference Guest Stream 67.1.
Andres Carada is here with us and will be presenting and discussing on NTQR, logic for
noisy AI algorithms, complete postulates, and logically consistent error correlations.
So thank you for joining Andres and also Jakub and looking forward to this discussion.
Thank you, Daniel. Thank you for introducing me. Hello, Jakub and Daniel. Jakub, would you
like to talk about why you're here?
Yes, sure. So hi, I'm Jakub. I am a researcher currently based in California.
I am interested in using physics-based principles to model intelligent and self-organizing systems
and how different frameworks, most specifically Active Inference, can be applied to both topics
in AI and systems engineering and broadly how we can use these formalisms to understand systems
at a deeper, more inclusive level with other disciplines as well.
And I'm very interested to hear your talk and your thoughts on these topics and engage in hopefully
a productive discussion.
Looking forward to that. Yes.
Okay. So I'm going to talk about the problem of self-regulation for any intelligent machine.
And it has been a long journey for me in dealing with this topic. It goes back to a patent that I took
out in 2010. But recently I've come to understand it because I've been collaborating with a philosopher
and an economist, the aspects of it. And that's why I have this very abstract title for it called
NTQR. Because it refers to a situation where you have an ensemble of N experts to which you have given
T tests. And each test has Q questions and each question has R responses. Okay. So it's about evaluating
noisy AI algorithms when you give them these types of testing protocols. And superficially at the
beginning, we can take that testing protocol to be the surface application itself. You are literally taking,
let's say, a multiple choice exam. But I'm going to eventually dislodge you from that to understand
that this is a digitizing format for testing logical consistency of anything. Right? Because you can
digitize anything. So even if it's continuous, right? You can create, you know, four response ranges,
right? And stuff like that. And the big breakthrough for me has been the recognition that for a long
time I talked about these things as universal thermometers. And the universality meant that they
could be used everywhere, just like thermometers can be used everywhere, aside from you melting them,
right? Or freezing them beyond their range. And the thermometer also had the notion of being stupid,
no intelligence, no theory about the world or the phenomena that it's measuring the temperature of.
And I've did that for a long time. And I took out a patent because I thought that I had found a method to do these
thermometers and you can patent methods. But it turns out that I did not discover a method or have a patent.
I discovered logical postulates for evaluation. And so what I'm going to talk about today is the logic of evaluating

noisy functions, period, in general, universally. And so I'm going to talk about them as postulates.

Because they apply whenever you are doing these NTQR tests. And so a major conceptual goal for me is to

convince you of that, right? That I have something that you could call postulates because, especially in the machine

learning world, people say, this is crazy, this cannot possibly be, right? How could you have postulates for anything in the real world so general?

And you'll see that I do that by basically shedding all representation.

So I want to mention these collaborators. One of them happens to be my cousin. This is a conflict of interest

disclosure. He's a professor of philosophy at Virginia Westland University. And he has been the severe critic of things.

And I, you know, I've done most of the work in terms of the mathematics and the machine learning.

But he's the person that's responsible for actually introducing the concept of logically consistent

versus logically sound, and the utility of having any sort of logical, right, proof for anything that you do.

And then the other collaborator with this paper is Ilya Parker, who works with the Navy and has a master's in economics.

Because you're going to see that this is a fundamental problem in economics that goes under the name of principal agent monitoring.

So I'm going to talk about, you know, the two main goals that I have in the talk, and I'm going to introduce the main

problem that we're going to be addressing, right, this principal agent monitoring problem.

And then I'm going to make some observations about you guys. I find that the format of these live streams is incredibly

interesting to me intellectually. And I just want to, you know, sort of give an outsider viewpoint about what I'm seeing that you guys are doing

that I find so laudable. And as I was saying to Daniel, I have a little bit of an imposter syndrome because I feel like you guys made a mistake

inviting a stranger like me so suddenly, but I hope to repay that kindness by engaging in a good discussion with everybody.

And so then I'm going to, you know, introduce this idea, right, of algebraic evaluation,

talking about binary classifiers, and I'm going to make the math be really simple, right, just linear algebra,

so that you can understand it, right. I don't want to lose you because this, this is eventually involves algebraic

geometry, and something called Grubner basis, and all these complicated stuff, but I don't want to use this

math to hide from the basic ideas that are behind it. And then I'm going to talk about what I said before,

that that you shouldn't think of the multiple choice exams as actually being the surface realization

of how something is being evaluated. You could, you could have given people a philosophy exam and

giving them written essays, and then you have chat GPT grade the written essays and give a score, right,

from, you know, zero to four, right. So then you're making an NTQR equals five tests, right. And so I'm

going to go over that. And then the main technical result of today, the main thing that is the, the

breakthrough is that there are complete postulates for pairs of correlated binary classifiers that allow you to

separate correlation from individual performance in a way that actually allows you to then immediately trivially sort of compute the only logically consistent error correlation, if you believe that they behaved in a particular way. And I'll go through that caveat. Then I'm going to show an exact solution for the tree of error independent classifiers, and then, then show some experiments, you know, and work, work through the math of doing it. Any questions or people want to comment or anything?
Go a little further, and I believe we'll have a few pieces to add in.
Okay. So the, the, the main goal that I have, right, is, is that I want to describe to you what I did technically, right, which is try to figure out how I'm going to evaluate, uh, ensembles of noisy binary classifiers. And so, you know, it has a particular way that I go about it, blah, blah, blah, blah.
And, you know, that's sort of like the surface realization. But once you realize how, how I've done it, and what I've been able to achieve, what I really want you to think about is this last question, right? If it's not this thing that I'm going to talk about, wow, isn't it something like this?
Because it seems like this is so easy to implement, and it's so useful if it existed.
And so that's where I want to take you. Okay. And so I want to have the discussion for that. And, you know, we can go into the math and I can pull up the papers, I can pull up GitHub repos, right? And I can show you stuff if you want me to show it, but I don't think it's going to help the discussion. And so I want to talk about, you know, sparsity, picking best variants, and safety from internal mistakes. That those are the topics. I think Jacob, uh, wrote, you know, talked about some of them, right? I think that the November 14th talk that I saw with, uh, Alexander, he was harping very much on sparsity, being a very biological thing. And you're going to see how I use it, right? Because I basically have a data streaming algorithm. Okay. So this is my outside review. I mean, this is a community of people that you guys have been working in so, such hard problems, and you're still working on such hard problems, which I have to say, you know, and I'm being, you know, honest here, you've made very little progress on, right? But it's because they're really hard fundamental problems, but you guys don't give up. You guys are still having this live stream. You're still inviting people. You're still discussing it, right? I love that, right? Not giving up on, on, on, on, on particularly hard problems, right? So I'm very attracted to that particular mind temperament. And you are a community, as I can observe, of people who have come to neuroscience and biological things, but from different places, sometimes from physics, sometimes from biology, sometimes from mathematics, and have arrived there, but you're still bringing in people from those places, right? And there's a lot of active listening that I saw Daniel writing stuff down, right? As people were speaking, I mean, incredible stuff that you rarely see. So kudos to you guys for setting up this sort of environment and, you know, inviting a stranger like me to talk to you.
It's very courageous intellectually, and I applaud it.
And I hope to repay the kindness by engaging in a discussion for the purposes of clarification, right? That's what we're here for.
Okay. So, you know, I, I, I, I think in LinkedIn now I have myself a scientist and inventor, and I'm starting, I think I'm going to flip it. I'm going to say inventor and scientist, because I'm, the older

I get, the more I think that invention is better than science. First of all, it's been around before us, right? And some people claim that the fire made us, right? And we didn't invent fire. It wasn't invented by homo sapiens. And so invention has been before humans, and it probably will be after humans are around, right? And, and so I've done, I worked in many different things. And so I, I've done, I have a patent for an underwater periscope. When you are at the bottom of a pool, when you look up, you see that, you see that circle, that's called snail's window. The whole world, the, the, the, the whole above surface world has been compressed into that circle and distorted, but you can see with the shading, right, that you can pick up pieces of the world. And so you could, if you have a polarized camera, you could reconstruct this. So it's this idea of reconstructing, right? What's beyond that I find it interesting. And I've worked in, in, I trained as a physicist, but I trained, you know, I, I've been always interested in the marriage of mathematics and exact stuff with a very physical problem. So for my PhD, I worked on superfluid film vortices on a torus, and I found that theorem of Riemann that had, had no application in quantum mechanics until this problem.

Okay, so let's get to the, to the meat of the problem. So the principal agent monitoring problem. So this is a problem as old as the hills. So it appears in Plato's Republic. And some people call it the allegory of the ship of fools, right? So there is a ship owner who has a crew of sailors, and you know, they're kind of rowdy sailors. Plato says that they may be drunk. And among them, there's a philosopher King who's Plato himself, right? And, and, but the owner, you know, doesn't know how to pick the sailors, right? He, the owner just owns a ship doesn't have any ability. That's why the owner has hired sailors. So if you are an a principal, and you hire somebody to do the job for you, either you're lazy or stupid, and you don't want to do it yourself. So you have this monitoring problem. How do you monitor, you know, how do you do, how do you monitor work that you don't want to do, or don't, or don't know how to do? Does everybody get that the paradox there?

Right. And so, and so Plato's answer is democracy is no way to do this, right? Because there can be factions, right? Part of the crew can have a faction, and you're not going to pick the best captain, if you have people vote for who their best captain is. So, you know, Plato used this to attack democracy. But this kind of goes, this is counterintuitive, right? We know that that group, right, that groups are better for us, right? That ensembles help us survive better, right? So there's this tension, Western civilization, between this fear of the mob, right, and this thing about the unitary right philosopher king that's going to be the dictator. So who's going to rule us? Is he going to be the philosopher king dictator, or is he going to be the mob? Right? So there's this constant tension.

So I came up with this problem, because I'm one of the hundreds of scientists that have made speech recognition possible on your phone right now. I used to work for Dragon Naturally Speaking, and Dragon Naturally Speaking, if you go into Wired Magazine and look at the highlights of machine translation, the founding of Dragon Systems in Newton, Massachusetts, in the 80s is considered to be a pivotal moment, because, you know, Dragon Systems went on to create the first continuous speech recognition product on the desktop. And the way that speech recognition was developed was, as people may know from machine learning, was by having data sets. And one of them was people reading the Wall Street Journal.

And then that being transcribed with all the mistakes that people made, and the whom's and the has, and so forth, right. But after, you know, five or six years, you know, the Wall Street Journal corpus became very important as a benchmark. And so people started realizing, well, you know, I'm being graded incorrectly in this transcription, but I actually, it's the official transcription that's wrong. And so why don't we just redo it? Right? And so they went through the whole corpus, and they redoed, they spent the money to do it better. And so they, so every test from then on, writing the software, had the old benchmark with the mistakes, so that you could still benchmark all papers, and then the new benchmark. And I said, who's checking the new benchmark?

Right? It's the principal agent monitoring problem, right? Who checks the experts? How do we know that this last transcription doesn't have any mistakes? Then a second moment in my professional life, I'm a lecturer at UMass Amherst in physics, right? Physics 101. I have to lecture 500 students. The only way to do this is to give them multiple choice exams, bubble sheet answers, right, that I then collect with the proctors at the end of the exam, and take to the back of the room and feed into an optical scanner.

I won't describe to you everything that I did, and I was told to do, to randomize the test in the room, okay? But it led me to sometimes when I was feeding those exams to say, why do I need to feed the first sheet, the answer key? Why can't I use the wisdom of the crowd to grade the exam?

And then finally, the thing that actually got me into the mathematics of actually doing this was in the in 2008, I started working in in in UMass Amherst in the computer science department with Howard Schultz doing digital elevation models from multiple maps that were done from aerial photographs. So how do you combine these things, right? There's there's so much data now, right? This is the problem that we have so much data. How do you combine it and not prevent the noise from corrupting it? And then finally, the 2010 patent by data engines on error independent model solution, which as I'm going to tell you has disappeared, the the patent is null and void.

There can be no patents on postulates, you cannot patent natural loss.

So I'm telling you the history of my business failure, basically.

Okay, so so this is the paper that was on, you know, the geometric let me get the laser pointer, the geometric precision error, right, for computer vision tests.

And basically, the idea there is that you're you have a bunch of maps, okay, and you're and I'm plotting here the error covariance matrix between the different maps. And the way that my boss, Howard Schultz did the maps, he did it so that he took advantage of the fact that when you have a photograph a photograph B, when you match features in a to be, you can go this way, right, one way an hour, but you can also go the other way. And that matching is not symmetric. So he wanted to take advantage of that non symmetry to discover blunders. And he did so and he was able to show that he got rid of blunders.

But he went on to continue using the maps. And everybody told him, Howard, why are you doing that? Those maps are really correlated, right? That's A to B and B to A matching. And he said,

yeah, but why not? They have some information. And so eventually, I was able to prove that he was right.
And this is what I'm showing here is is where I have a model where I just assume that everybody's
zero and all and I've induced a signal if you're a physicist, you know what I'm talking about,
right? Because I have these two pair to these two maps that I know have been correlated or highly
correlated, and I'm introducing them there with other maps that are not so correlated.
So it's like, I mean, I'm putting an injection, right, a modulating signal. And this recovery here
was with compressed sensing. And you see how without being told that that along the diagonal,
there's a strong structure, it's able to pick it up. Right? So sparsity, if the errors are sparse
enough, you can discover the error, the precision error between regressors. So even though today,
I will talk mostly about classification, everything applies to regression too. And here's the crux of the
issue, right? That the way to do this is to basically get rid of reality. This is the way that
you get rid of representation. Right? If I have a model, why don't I subtract the true value from
the model, whatever remains is the error. And so I'm just going to talk about that today in terms of
classification. Okay, we can stop here if people want to ask questions, because the next theme is
going to be data streaming. Yeah, I have a few comments, Jakob, you want to go first, though?
Yeah, sure. I guess. I was wondering if there, if you have been thinking about,
I guess, a measure to quantify how much information is lost or can be lost when using a particular benchmark.
Because the way it sounded like when you were talking about it is, I guess, similar to the notion
of, well, all models are a coarse graining of the particular thing we're modeling. Therefore,
every benchmark is going to have this problem of who's watching the benchmark.
or who's checking that the benchmark itself is correct. Which is a, I suppose...
It's an inescapable problem. We cannot escape it as a civilization, right? Nobody knows what the answer
key is. Right? There is no dictator that we can go to as a civilization, as a society. It's inescapable.
Right? We're always benchmarking to something else. There is nothing but benchmarking to something else.
Right? Am I crazy here? Or can I get an almond? Right?
Nothing but difference? Yes. Nothing but gradient? Unity is plural? Yes. At minimum, too.
Yes. Yes. Right? That is the only thing that we have to observe. And that's a statement of
relativity in physics, too. Right? That we cannot observe absolute velocity. We can only observe
relative velocity between particles. It's the same thing for measurement. It's no different.
Yeah. A few other comments. So the back and forth, keeping the back and forth directions being fit,
and how that contains information on blunders, like you described. And then another subtle pattern is in
the matrix one slide previously. In the cells that were two-off diagonal, yep, there's a slight positive
correlation that's transmitted spuriously, potentially amplified through compressed sensing and distorted and
so on. There's also ripples of underlying even trivially analytical square correlation matrices
empirically plus noise plus compressed sensing have these attributes. And that is what puts so much of
the experimental design and attention interpretation onus onto the cognitive entity as hardly this or any
data speak for itself. Yes. Yes. Yes. I have to agree with you. Right? There is, we have to

distinguish here, there is no magic and there's no metaphysics here. There is only a measurement, which is a statistic of a sample. And therefore it's impossible for it to, for you to generalize it.
Right? It's not an, it's not an intrinsic property here. Right? If I'm telling you that this is the error correlation in those maps, I'm telling you that that's the error correlation for that particular collection and for those particular maps, not for the algorithm in the future in the, or in the past.
Right? I'm telling you something about the test that you just took period, end of story, nothing else.
If you want to ascribe meaning to that, go ahead. Right? But then you're incurring that, right? That cognitive, you're the one who's now becoming the dictator, right? The bucket is stopping with you because you've decided to assume that that measurement meant X about the world.
Yes. When you undertake the epistemic quest, then you're the principal epistemic agent. And that's an n equals one situation. Another, I think, interesting touch point is, um, hierarchical predictive processing models, which we talk about a lot in inactive inference, they have a highest level prior.
And if that is going to be also learned over updated, there's going to be a hyper prior on that.
So that's one aspect where there's just the buck stopping at a point from a hierarchy perspective.
And then also, um, even with just the Gaussian, the bell curve with the mean and the variance, but there's not a variance on the variance. That is what gets collapsed. That's the metacognitive. And that degree of freedom is what constrains and makes models go stale and makes them overgeneralize without being aware. Yes. And I want to, I'm going to address that one here because I'm going to talk about measuring error correlation independent of model, being able to measure empirical error correlation. Now, whether you want to generalize that because you have some view of how to base and generalize it or not, that's a separate issue, right? But being able to at least benchmark the, the actual correlation on the test so that then you can have those cognitive models and those epistectic models that can have that. Let's make the analogy. Your car has a computer in its engine, right? And it also has a thermometer. What I'm going to talk about today is about how to build a thermometer that helps the car's computer run the engine better, right? And I'm just, all I'm saying really comes down to that, that it, wouldn't it be great if intelligent minds had a thermometer for intelligence, right? Because then the thermometer just tells you that the car is overheating. It doesn't tell you why you need a cognitive process, right? To figure that out. Oh, it's because a piston is blowing, right? Or the oil went down, right? That's a separate issue, right? But knowing that something is wrong by overheating with a thermometer is golden, right? And I want to get to that rock of having a thermometer that I don't have to worry about representation. It just gives me a measurement. I'll deal with the fact of what it means, right? But I just don't want to worry about what that measurement is, or I want to reduce it as much as possible. So that epist, so it may sound weird, but you would want a tool that strips epistemology from checking the logical consistency of your evaluation.
So that you can aid that epistemic agent. And so I'm right, this is the thing that I find weird and fascinating here, that I'm going to propose to you that what we need to do is strip epistemology. And I'm going to do that, you know, continuing on. And so let me now start talking about that, okay? So data streaming. So you guys talked about sparsity, right? So the way that I encountered

data streaming was with the good Turing smoothing algorithm in speech recognition. At Dragon Systems, they gave us a book written by, gosh, how can I forget his name? Our research director. And he had good Turing smoothing, frequency smoothing. Can I get a count of how many people know about what this is, or know what data streaming is? Is this something that you guys know very well as a community? Turing smoothing. Give a description that sets it up for how you want to talk about it.

Okay. So the way that the typical data streaming algorithm and the way that it gets presented is that it's a method for minimizing memory. When you want to compute statistics of a stream. And so the prototypical simplest cases, you have a stream of numbers that are coming down the the pike, right? And you want to keep, and at any moment, I want you to tell me what the average of those numbers are. So the memory intensive way to do it is to store all the numbers, right? And whenever a query gets made, take the average of those numbers, right? The whole list. But of course, as the stream increases, right, that list is going to balloon and become humongous. So the philosophy of data streaming is is don't do that. Just keep two numbers. Have a sum. So every time you get a number, increment the sum and then have a counter of how many numbers you've seen. And then the average is just the sum divided by the number. Right? So just having two numbers, two integer counters or two real numbers, whatever, right? You can store and at any time compute a stream that may be a billion long.

So that's the purpose of data streaming, right? To compress, you take a data sketch, as it's called, which is a compressed version of all the events that you've observed in the data stream in all the richness, but you're not doing that. You're just taking the events and you're doing some sort of compression on it. And you're keeping that statistic and you're, and then at the end, when people tell you, ask you, you say, oh, this is the average that I've served so far.

So for good touring smoothing got invented by Alan Turing during World War II, because I don't know if you know that he was responsible for the Enigma machine breaking, right? In World War II. And, you know, they had, they started to, you know, Alan Turing was a very good mechanical dude. He actually built a tide predicting machine way before computers, or he did any of the stuff that he was doing, you know, as is known very abstractly, right? He was very much hands-on and he helped build a bomb, which were early computers that were processing the German traffic that day and trying to decode it.

They had to work all day. These bombs were continually churning, right? This is the Enigma machine, by the way, recently found in a flea market, $15,000 if you find this machine. So it had these very complicated rotors, right? And so these, these are the bombs and they had a lot of women assistants, right? Back when, when computers were women, right? Before you had the Grace Allen and so forth, right? And so this, you had to send people home at the end of the day. When did you stop? You don't know how many secret rotor settings were done that day. When do you stop? When do you decide that you've done enough processing, and there's no big chunk of German traffic that you've missed? So he came up with good touring smoothing, which was how many rotors settings have we not seen?

Which is a crazy statistic if you think about it, right? What do you mean? How many things have you not seen?

Right? And so he was able to do it by shifting, right? By saying, well, there's many rotors for
which we've seen only one message, right? And there's fewer rotors settings for which we've seen
two messages, right? And so if you look at how many messages were per rotor setting, which now biologists use to figure out how many species they miss in a survey, you can then know how to shift the mass to the zero
frequency observation. You take a little bit of probability from everything and shift it down.
And can you believe that this is still one of the provable best ways of doing this? Estimating unknown
frequencies? Alan Orlarsky from UC San Diego has been doing some of the modern work on understanding
good touring smoothing. Notably, it doesn't get used nowadays in neural networks because neural networks
never have any missing data. They will always give you an answer, even if it's wrong.
Or Linske in this paper, which was published in NeurIPS, argues that that's wrong and you should go
back to doing what it does, but we're not going to go into that.
Okay.
Can I make a comment there?
Yes.
Linske in this.
Um, this reminds me of the distinction with parametric and non-parametric statistics.
And when people talk about how many parameters this model has neural network transformer,
what have you, it's a parametric model. That's why they're talking about how many parameters there are
and non-parametric methods. Although that's kind of like saying the non elephant animals kind of classic
joke, it's also like a very open and procedural space where there can be a lot of methods such as
shuffling and leaving one out, all these different statistical methods that may have less power or less
of some statistical positive capacity. In the case where the generated data, for example, are known to
be from a Gaussian error distribution, but under relaxed assumptions about variant structure,
then non-parametric statistics can do well.
Yes. And in any end, and heuristics must be something that biological systems use a lot.
Right. Instead of exact computation.
I would my toast that statement into two, which the first statement I'm most comfortable with is we
can model biological agents as using heuristics. We can model anything as using heuristics,
in our finiteness as cognitive modelers. So if we say we're restricting ourselves to an eight
parameter model or to this kind of genre of model for this phenomena, it's a map territory distinction.
Now, as to whether the territory uses heuristics, whether you can really say, does a bacterium or
does a squirrel use heuristics? That is a question. What are those heuristics? But the first question,
which I feel most strongly about, is that we can say that we'll choose to use heuristics in modeling
complex behavior. Why not? How can that be not the case?
Okay. I'll grant you that.
Because I don't have actual experimental, right? I mean, to be precise, right?
Scientifically, you're asking for a particular demonstration of heuristics in a biological

system, and I don't have that. I only have an aspirational statement that maybe does happen, right? Okay. So the thing that I want to do here and the problem that I'm trying to solve, again, is not necessarily procedurally what's happening. We know that us in the world are not completely isolated from the world, right? But what I want to think about is the gedanken extreme, where you're isolated and paranoid, which does occur in certain technological realms. It doesn't have to be a mental paranoia, right? So for example, a self-driving car that wants to check its pedestrian detectors immediately, right? That immediacy, right, means isolation, right? It can't wait to get the feedback by actually touching something, right? Or get further information moving forward. And, you know, it's close, right? It can only rely on its own things because it's unlike a Roomba, right? It can't be bumping into pedestrians. It cannot get feedback any other way, other than this mechanism that it's decided to do whatever it is for pedestrian detection, which could be sonar or could be visual.
So I think of it as an extreme that I want to solve to understand what are the limits of what can be done, okay? Not because we are actually need to be in that realm. And so I wanted to build certain features into it, right? Because I was using this in speech recognition. And one of the things that I wanted to do was, you know, to make it simple. And so I wanted to treat all the members of the example as black boxes. So I don't want to go inside your brain. I don't want to go in and get readings of internal parameters in models, which is not wrong, mind you, again, right? To come back to the analogy of the thermometer and the computer in a car engine, right? You can have intelligent things. And a lot of people do neural networks, right? And they inspect internals to try to figure out, right, how to detect when it's actually hallucinating. There's nothing wrong with that, right? But I'm saying that it itself is then going to bring assumptions that I can't check, right? So I don't want to do that. I want to treat everything as a black box. I just want to see what the decisions are from the oracle, or whether that oracle be a person, a function, or whatever, right? And I don't want to go, I don't want a theory of mind. The other thing that I want to do, and here's the crucial thing, right, is I want finite chains of safety validation. I don't want to use Bayesian models.
I don't want to use probability. Because like you just mentioned, Daniel, those are hyperparameters, which are encoding information about the world. And the whole point about being paranoid is that I cannot, right, I cannot rely on that, right? I cannot rely on any representational knowledge.
I don't want to, right, to avoid that pitfall.
And so by going to sample statistics, that's how I'm going to avoid that.
Because I'm not going to talk about distributions. I'm not going to infer anything about the future or the past. I'm just going to tell you a number about something that already happened.
And since it's a sample statistics, right, I'll be able to then do the magic that I do, which is to create a complete set of postulates. So then I can prove that you could be in states that are probably wrong or inconsistent. And so that's the punchline, right? That I can do both of these things of being isolated and gedanken, right, and paranoid by just having functions that only use statistics of the decisions by the ensemble members, nothing else.
So let's stop here because this is an important point.

Right? This is a design choice, right, that I'm making.

Could you unpack that a little further? I'm not sure I fully understand how using sample statistics

differs conceptually from using, say, Bayesian probability on your data.

So would you agree that there's the concept of sample statistics and then, you know, distribution statistics,

right, and things like that, right? That you can always define a sample statistics and actually compute it

if you knew what the sample was. So that's what I'm talking about. That's what I mean by sample statistics.

And then I mean by, so what would it be? It would be something like if you have data being produced by a Gaussian distribution,

right, a sample statistic would be the mean of what was produced, okay? That has meaning, that exists, right,

independent of what meaning you want to ascribe to it. Whether you want to think that it's the mean of the actual Gaussian distribution, right?

That's a separate thing, right? Using that data and using that mean to then estimate what is the mean of the Gaussian distribution that produced it is a complete different task.

Yes? Right. So for one, the data may not have been produced by a Gaussian, right?

This issue, just to connect it to like what we talk about with variational or just Bayesian inference and active inference,

let's just say in the room, the temperature is continuously variable. But then the thermometer is going to be integers.

And then the agent is only capable of coarse graining into buckets of 10.

So we can still talk about...

Yes, only cares because of, you know, a precision engineering spec, right?

Why am I being uselessly precise, right?

Yeah. Yeah. Variable precision to promote system properties is definitely a corollary of this kind of a framework.

More broadly, like getting rid of the representation question.

This is like hierarchical predictive processing of visual.

And so at the earlier lower levels, and this phenomenon is recapitulated in silico, as well as in aspects of biological systems,

that the earlier layers pick out less abstract components like edges and on and off.

And then not that it's clean, but that at each synaptic interval, which can be modeled as actually means and variances and differentials in the hierarchical predictive processing architecture,

that can be modeled as like a kind of like representation explained away.

You know, where's the rom-com in this pattern of neural activity?

And then it's just very interesting that you brought up, of course, the engineering setting.

Predictive coding, which is to say difference in coding, was discovered slash invented in the video compression setting.

And yet in the Rau and Ballard 1999 work and so on, it came to be understood as a more general property of how information transmission and so on,

in certain settings led to certain parsimonies that has certain relationships with things like the Kalman filter and predictive coding.

So, you and Jacob have mentioned this thing about transmission of information.

So, I'm not going to go into it. There's so much to talk about. I've been working on this for a decade.

The difference between the observed responses gives you something about the test,

but also something about the quality of the person who answered the test.

The uninformative person who answered the test doesn't transmit anything about how they performed.

The random guesser only transmits information about the test.

If I randomly guess what A and B are, right, I'm going to have the right proportion of A and B as the test itself.

It's going to be in the wrong places.

Yeah.

Right. But, but, but you're, but you're transmitting the difference between the A's and the B's.

And so there is no magic here.

If the ensemble is made of stupid agents, they are not able to transmit via their aligned responses, any information that allows you to grade them.

So my method has blind spots.

Right. There is no magic here, right?

If your ensemble is made up of stupid members that don't have any information, you're not going to get anywhere.

What do you think about an ant colony where none of the nest mates know about how many seeds that they have and how many nest mates need what and how much rain there is outside and the rate of information and blah, blah, blah.

But they don't know any of that.

They're just using their local interactions.

I actually have a couple of books on ants and ant colonies.

And I saw that you wrote a paper on ant colony behavior.

I'm very interested in thinking about that, but we're not going to be able to talk about that.

Okay.

Fair enough.

But, okay.

So I'm interested in the comment you made about route to general information.

So data streams as NTQRs tests, right?

So we're going to look at binary classification, right?

So back down here below, we have the true label, right?

So this is a stream.

So think of this flowing to the right, that gray box.

Everybody sees that alpha and beta?

And then I have three different classifiers that are making different decisions, right?

So aligned on the first item, the first two classifiers are saying that it's beta and the third one disagrees and says that it's alpha.

Everybody with me on this as being, right?

And so what I'm going to do is just collect how many times each of these different patterns occurs.

That's the compression, right?

So we could be looking at a test that has a million items.

I don't care.

Since I'm doing binary classification, I only need two to the three, eight integers.

So right there, we see the compression that's very attractive, right, for in terms of a biological system.

Because you're not storing the whole stream.

You're just storing these eight integers that are picking up these patterns.

Okay?

And like I said, and like Daniel has said, right, you can digitize to whatever you want in terms of engineering spec.

You don't want to measure the degree of the temperature of a car to a millionth of a degree.

It's useless, right?

Economically, there's no point.

Likewise, you may not care too much to worry about, you know, knowing, right?

You don't care about too much beyond boiling or something like that, right?

You just want to be told, hey, the car is overheating, right?

You could digitize that.

That's a safety.

You're coding in there by choosing the degradation of digitization.

You're deciding what is your engineering spec and what is your safety concern, okay?

So I cannot tell you that.

This is an instrument that you have to design.

And you have to decide what your R is.

Any questions on that?

Yeah, just a few notes.

This is really interesting.

There's the principle of the hash encoding or the kind of lookup table for all the combinations.

And then you can do hierarchical nestings.

And then also this situation is kind of like the cells in the retina, just speaking coarsely,

that there's some false positive and false negative for some activation thresholds.

And so it's kind of like a retinal display that is getting noisy photons.

And also there's variability through all of these finite systems.

Variability and finiteness are ever prevalent.

But the observation is just what it is.

Exactly.

Exactly.

And therefore, right, all the problems with philosophy and logic occur because of infinity, right?

If we just accepted that the world is finite, but maybe very large, right, we would solve all sorts of problems, right?

Zermelo's axiom of choice is only needed for infinite sets, right?

Finite sets, everything is explainable and understandable.

Okay.

Okay.

So here I'm going to show you, you know, eventually you're going to see this, this, this, this is the set of equations for three classifiers.

And these are the eight frequencies, okay?

And so these are the complete postulates for error independent classifiers.

So I'm going to come back to this and make it much simpler than it is, right?

But I just want to show you how, you know, these frequencies here, right?

There are eight of them and I can sort of synthesize, right?

And assuming that, that, that, that the classifiers have some sort of performance, right?

If I just, I randomly pick numbers between one and a hundred and then divided by a hundred to, to have these values.

And then I just plug them back in here and then I get what, what the frequencies would have been of how they voted.

So I, I'm just trying to show you how these numbers, right?

Are integers, right?

There's no probability here.

There's no probability here.

There's no probability here.

And then the other thing I want to go back to is, is something that I think you guys would find interesting from the physicists.

Renaud was a 19th century French physicist, very famous for building the most precise thermometers of his time.

At the time when people were just inventing the concept of temperature and people told him, how do you know that you're measuring anything?

Temperature is an imaginary concept, dude.

We don't even believe that there's such a thing as temperature.

And even if you did, you would need to have some sort of theory about how you make errors in your thermometer so that I would believe what your thermometer is saying.

And Renaud was, screw that, right?

He says, are you saying that there is no science without theory, that there can be no purely empirical measurement?

And he said, no, I'm going to make purely empirical science.

And so he was a radical empiricist.

Eventually in the 20th century, right?

We found out that empiricism doesn't work.

But he basically is going to do what I'm going to show you today.

And this is what I did with the maps.

He took thermometers and just compared them against each other.

That's all he did.

And then he looked at the differences, right?

And he's the first one to start talking about the concept of precision as saying, yes, I can build the most precise thermometers because I can get them to agree to certain amount of figures.

Right?

And then they start disagreeing.

So even in science, right?

There's a concept that you really do not have science until you have disagreement in measurement.

You have to have error in measurement.

If you do not have error in measurement, you either tell me about math and postulates or you don't know what you're doing.

Your instruments are stuck.

You think they have a precision that they actually don't have.

Yes.

Thank you.

This is awesome.

When we have multiple observations going wide, that's sometimes called sensor fusion.

And then when we have the clash of the prediction and the observation and the differential that's found there, that's the predictive processing architecture.

So those are both, there's so many motifs, but you totally need a minimum of two.

Otherwise you have like a statistical monad, which is really just a pure conceptual object.

And then one interesting tie in is like in active inference, as we specify the generative model, the preferences are directly over observations.

Like the, in extreme case, meaning that the likeliest thing and the preferred thing are one in the same.

And it's just that it just, you prefer to get that number on the roulette table.

And every single time your action is confirming, that's the case.

That, that, that is one special case that can be set up as well as epistemic, more general epistemic activities.

But by having the pragmatic value load directly onto observations, like the thermometer reading, not I have a preference for the room to be at 72, but I have a preference for something about the measurements.

And that direct target rather than the hidden state being the direct target, which leads to a fantastical analysis.

So, okay.

So, okay.

So, you guys seem to be, yeah.

Yes.

A head of, about thinking about this and how to implement it.

Okay.

So, let me finish with this, with the temperature saying that, that the reason I came up with, with upon Renault was because that this thing happens in invention all the time, right?

You're not too clever.

You invent something and you think you're clever, but you're not, right?

It's somebody's thought about it before.

You just don't know it, right?

You're just ignorant.

And so I said, you know, somebody must have thought about this.

Somebody must have done something like this.

And so I went looking for it.

And so I found this book, Inventing Temperature, which has a whole discussion about Renault and how he's wrong because he thought that he was doing a metaphysics, right?

He thought that he was getting behind the curtain.

That, that was his philosophical mistake, right?

And so I want to pull back from his philosophical mistake and say, no, this is the comment I made earlier to Daniel, right?

I have, I have, I am not saying anything about what the reality is.

I'm only telling you an estimate of a sample statistic.

That's it.

How you want to interpret that is up to you.

And I recommend this book because it's great reading, right?

Because it's about how you invent physical concepts, right?

And, and, and the struggle that people went.

Okay.

So now we're getting into the, the, the, the, the part where I think you guys are going to be interested because I'm going to back, go back to the digitizing the format for logical consistency, because there's two different viewpoints of the same test.

And one of them, which is the one that I'm choosing to use the NTQR methodology, just to have a new word for you to go somewhere else in your mind is different from the ML binary classification case, which has semantics.

And here's where I want to show to you that the mathematics of the NTQR test is separate from the semantics.

It can be attached to semantics, but they're not the same thing.

So I want to spend some time to make sure that we capture this one.

Okay.

Okay.

So, and then the, the payoff here is the, the realization that you can start to understand it in terms of semantics.

This is something that a machine could use to binary classify things in the environment.

Okay.

Which is how I thought of it first and how people in machine learning think of it.

Right.

I have a binary classifier because I'm looking to classify things in the external world.

I'm looking to be told how many A's and B's there are in the external world.

But once I pulled the rug from under you, I'm going to tell you that the exactly the same mathematics can be used to figure out the statistics of correctness.

And that has nothing to do with what A and B means in the world.

Because it just means that there's a, some percentage of the questions have A as the right answer and B as the right answer.

But who knows where you came from digitizing that format, right?

So there doesn't need to be any semantics attached to A and B.

And I can still calculate what your performance in the total test is.

So something that can be attached semantically then has the same mathematics in a way that it can be used to check tests about chemistry, about philosophy, about geology, about any subject.

Right.

You're not doing binary classification.

You're just correcting grades that a test that have two responses per question or three or four.

That's a general thing.

Yes.

Very separate from the specific task of binary classifying things in the world.

So I'm going to stop and, and, and, and, and encourage discussion because I want you to get, I'm going to show you the differences too, and how it comes into the mathematics.

But I want to first get, if you, if just with words, you understand that there's binary classification where the percentage of A's and B's in the test actually means something in the world.

And then there is grading of multiple choice exams where the percentage of A and B's has no meaning outside of the test.

It just means that, that 10 questions in the exam had A as a correct and you know, 10 had B.

But you know, the questions could have been about geology, right?

A and B doesn't mean anything.

Yeah.

Just to kind of say it how I'm seeing it.

If we look at the output of a statistical test, we might get a vector percentage rain and percentage not rain.

Bayesian causal model, probabilistic statistical model, percent dog, percent cat.

In contrast, we have a different procedural nonparametric approach that can be designed incrementally to discard various aspects of information,

especially in a discrete finite space.

Taking kind of the finitest perspective on modeling and then using that to say, well, I am coarse graining into this many cells.

So I am going to have this many hash codes.

So we will be able to do this in this runtime.

Okay.

That's a little bit too complicated, Daniel.

Let me, let me restate the following.

We have a philosophy question.

Okay.

A philosophy exam.

And, and the, and the, and the philosophy exam consists of, you know, that A and B questions, right?

Where you giving a passage and you're giving two philosophers and you have to identify which philosopher wrote that passage.

Okay.

You get the philosophy exam.

Yeah.

It's testing for familiarity with philosophers, not the ability to generate philosophy or recognize new kinds, but I hear you.

Right.

But, but then, so, so, so we could have, you know, Schopenhauer and Wittgenstein as choices A and B in question one.

Right.

But question two could have Plato and Aristotle.

So A and B, right.

Are not referring to the same thing in the two different questions.

That's what I mean by having no semantic connection.

Right.

Right.

But your grade is that percentage of times that you correctly answered A when it was an A type question.

Right.

So is that, is that make it easier to understand that I'm making a simpler point, right?

That I can detach the grading you from having, attaching a semantic meaning to your response.

It's, it's like you, this is a common question in, in neuroscience where certain cell, certain cells while learning a task can learn an abstraction of, of the task.

For instance, if it is A or B, you will have cells that respond to this concept of A or B, but then you have specific components of, of the network, which specialize to the particular instance of A or B like Aristotle versus Plato.

Um, or any other, uh, uh, set of, um, set of philosophers.

I guess the question is whether, um, I'm not sure I fully understand the, the, what, whether.

So both can happen at the same time.

Right.

So both can happen at the same time.

Right.

You recognize that, that you can have the general structure, right.

Where, where basically if, if, what I'm trying to argue is, is that if there's an algorithm that can grade art.

Art equal to questions, right.

Then I'm done.

Right.

It can, it can do it for any article to question or philosophy or economics.

And it can also do it for binary classification.

What do you think about binary classification of language, like language generation questions?

Yeah.

Any, anything.

But, but you would have to decide if that's correct.

Let me, let me go on.

Maybe, uh, you know, and then, uh, when I show you things, you, you can start, uh, you know, I'm not sure.

But, uh, let me see if I can, let me escape and see if I can go now to Mathematica.

Oh, so I get to show you some pictures.

Yeah.

Yeah.

Yeah.

Like the example.

Yeah.

Like the example.

Yeah.

I think of the pheromone perception in the environment and the alarm or no alarm.

Is it at the threshold?

You got finite number of antenna, finite number of receptors, noisy signaling, go, no, go.

Buck stopping with an estimate brain somehow, some way.

Okay.

So let, let, let me go on and, and you'll see what I, that, uh, I think you'll start to see
how there's different ways of looking at things.

We still only see PowerPoint.

Oh, sorry.

I see.

Yeah.

I'm going to do share again and I'm doing share now of Mathematica.

Okay.

You see the Mathematica?

Oops.

Not yet.

I just took it away.

I don't know why I did that.

You see it now?

Yes.

Okay.

And then, oh, I'm going to stop the show because I need to do that.

Yeah.

Thank you.

Yeah.

That way I avoid.

Okay.

Perfect.

Okay.

And then now I can go to here.

Yes.

Perfect.

So this is the paper that I sent Daniel and it's the one that I've submitted to, uh, a special issue on AI safety and philosophical studies.

And so I want to, you know, so, so because I'm doing AI safety, right?

I want to, I want to speak about this specific case of doing a binary classifier to be concrete.

Right.

And so I'm now shifting into, uh, getting you into the mode of accepting that in fact, they are, there's a logic to evaluation.

That there are postulates and there's a logic and that it's universal.

Right.

It's a big one.

It's a big burger.

And so the way that I'm going to do it is by just, you know, talking to you about how I would, what are the possible grades on a, on an exam that has two responses, uh, before I see anything.

Right.

And so I know that if I have, you know, Q, uh, you know, a Q, a questions, right?

Of a type, then the number of correct responses.

So R is for response, right?

On the a questions that the I th classifier, right?

Does everybody see that notation?

R a.

So by right, this has to be, these relationships have to hold.

I claim these are postulates.

I, you know, I've been thinking how the hell did Euclid do this to begin with?

How do you introduce postulates?

What do you argue?

What do you say?

Right.

When you have postulates, you can't prove them, right?

You have to sort of say, I can sort of prove them here, but you have to sort of see them and say, yeah, of course, that makes sense.

And this, right.

You can't have more correct responses of a than there are a questions or B.

Agree.

Agree.

Or within a circumscribed axiomatic verified information environment.

So play on where I have Q questions, right?

Where I've told you what the number of Q questions are, right?

There's no way there's no, there's no other way.

Right.

And so therefore I can, I can immediately go and write, you know, this, this table.

I mean, this is how simple it is, right?

I'm going to tell you here.

Let me, let me make that bigger.

Right.

Right.

So you see this, I'm having a Q equal 20 exam, right?

20 question exam.

And I'm just going to make a table of all the possible correct A responses you could have.

A, A, right?

The number that you, of A's that you correctly added, said that it was A.

And the number of B's that you correctly said were B's.

And then I'm just going to keep track of the number of A's, right?

Questions.

Well, the number of A questions can only go from zero to the total questions, right?

And then once I decide that there's N A of them, then like I said, right, my correct responses can only go from zero up to that one.

And then if I have any questions, the B questions has to be this remnant.

And so if you just do that, this is, this is the cube you get.

This is what you get for the grades.

Until three weeks ago, I've never seen this cube.

And it's amazing that this cube is, it's actually, you know, kind of a nice figure, right?

It has got this double edged kind of thing, right?

But that's it.

If you answer a 20 question exam with 20 responses, your correct grade is somewhere, is a point somewhere here.

Is it a true tetrahedra?

I don't know what to call it, right?

It's got that 90 degree twist, right?

At the bottom is 90 degrees to the top.

Yeah.

Well, wow.

What do you see in it?

What does it mean to you?

Well, the thing about it that, that, that I found really interesting is let me show you the other way of looking at it.

Right.

Which is now the way that I looked at it for years until this year.

Which is the, the reality in which I looked at it, which was the, the, the machine learning.

Where now I'm telling you what is the percentage of things that A's that you got correct.

I'm telling you the percentage of B's.

And then I'm telling you the percentage of, of A's.

And in that cube, that same Q equal 20 test.

This is how it looks.

See how different that is?

See how it has that kind of quadratic structure.

What's different between the two.

Okay.

So I'm going to show you what's different is that I decided to do the math in the machine learning world.

Where, so, so this is what I'm, I'm sort of showing you now, right?

These are postulates, right?

The number of A responses that you give me are equal to the number of A responses that you got correct.

Plus the A, the B's that you incorrectly label as A's.

Yeah.

True positives and false positives.

Exactly.

Exactly.

Right.

This is, these are postulates, right?

These are prior postulates, right?

They, they exist before you see any observation about the test.

Right.

Everybody's on board with the fact that this is postulates, right?

And the, and that the number of responses you give, right?

In a sort of, uh, election integrity, right?

There can be no more votes than their voters.

It obeys this.

And I can rewrite this in terms of being a binary classification with these equations down here.

Yeah.

Just to kind of connect that to the case of an election.

So it's like one approach would be to take in the samples as evidence and try to estimate a continuous hidden state.

And you're always going to get into this statistical approach versus treating the fundamental finiteness.

We had, I'm not going to use the NTQR, but those are the parameters that describe the finite printing and dissemination project.

Yeah.

Of that project and that printing, there was a thousand ballots with five options.

That is the state space.

You could go into cognitive modeling.

You could do narrative information.

You could do interpretation.

You can have this, all these things, but the finiteness and the discreteness.

Then to get around this challenge of like, well, temperature, like in, it could be like any number, or it could be a number greater than zero.

So how do you deal with that?

And you take a kind of engineering approach here, which is you define the safety zone, which can be hierarchical, abstract, safety zone, generalized anomalies, and so on.

But you define the safety zone.

Within the safety zone, you have the direction of free energy minimization, statistically control, or you just stay in the fully finite world.

And maybe the issue can be resolved by explaining away the semantics of the decision through mere propagation and procedure.

Exactly.

Logical consistency.

Okay.

So, so, so it turns out that the, that, so, so these two postulates, right?

These two postulates here are written as a binary classifier here, and you see that these equations are linear here, right?

So that's the cube that you saw with the planes.

And you saw that when I went to the ML, I showed you parabolas.

That's the quadratic.

Are people seeing that in the equations, right?

That when I change to the ML point of view where I'm dividing, right?

Then I end up having these quadratic structures, right?

In the, in the cube.

Right?

You see the parabola?

What does that represent to you?

That's the ML space.

That's where I'm telling you the accuracy as percentages.

You see?

The percentage of, you got half.

Yeah.

It has the, this, this test has the highest resolution.

When the coin is even 50, 50 statistics does the best job.

When the coin is 99 and one, then three measurements give you very poor statistical power because you can't resolve 99, one from like 98, two very well.

It's not quite like that, but it's close to that.

But let's move on.

Okay.

So, so, so this, so, so I'm not going to show you the math, but these two equations, right?

This is where the algebraic geometry comes from.

This is what we can observe, right?

We observe how many times you said that it was a, we can observe how many times it was b, that you said that it was b.

And I know it's related to your individual performance by these, this, these equations.

It turns out that you can disentangle them.

And there's actually, you know, because these two frequencies have to add up to one.

There's only one postulate.

And that postulate is being represented here in the two different ways.

It's either this one or this one, either in the ML world or the, the NTQR world.

And here's where we come back to that thing about transmitting information.

The difference between your responses is telling me something about the difference in your correctness.

Just to clarify, like you use a single equation number 16, you're.

It's exactly the same.

It's exactly the same test, but I'm showing, you know, P of A sub A is RAA divided by QA.

So are you co-asserting these axiomatically or are you declaring them equivalent?

They're equivalent.

And so they're co-asserted by equivalence.

Yes.

And so why do the shapes look different then?

Because they're in different spaces.

One is in the P space and one is in the, in the integer space, right?

One is in the integer ratios and the other one is in the, the whole integer.

Okay.

You got it.

Yes.

So it's kind of like a discrete mesh and it's continuous space and there's integrity between the discrete and the continuous.

And you have very strong procedural guarantees on the discrete.

And then you have arbitrary or standard statistical guarantees on the continuous.

But my problem is that I have difficulty seeing how a biological system could do the division one, but I see, I can see how it could do the integer one very well.

Right.

That's why I'm bringing up the NTQR.

That's why I think you guys would be interested in it because the ML space is not, I don't see how a biological system could be dividing one, but I can see how a biological system could build a cube.

That's 20 sided.

Yeah.

That's kind of like a synapse and the synaptic release it's in vesicles.

Like it's not like dopamine is a continuous variable at the synaptic level.

The dopamine release is in one bucket.

Yeah.

Yeah.

So then, so, so then we get to the, so, so this is the, the, the, the interesting part, right?

So, so what is the problem with ensembles?

Gosh, if we could just do my, so this is where we go back to heuristics, right?

And my love of heuristics.

Majority voting is such a great algorithm.

It's so simple, right?

It's so simple and it works.

If, if, if things were good enough, majority voting would just be good enough, right?

It would just make us safe as it is.

The problem with majority voting is that it can go wrong, right?

The mob can be wrong, right?

Things could be flipped, right?

In terms of their accuracy.

So we need to consider now ensembles, right?

And, and, and the problem with ensembles is that they can be error correlated, right?

They're not going to be making their, their errors independently.

So we need to handle that.

And so here's where we come to the first set of what I would call non-trivial postulates, right?

Because this is what makes them complete, right?

That if I observe two classifiers, this, right?

They can only vote four different ways.

And there's no other way.

That's it.

That's the completeness.

This is guaranteed.

So if I give you equations that describe these four different frequencies, and those are going to be the four postulates, I'm done.

And so, by the way, these are the surfaces for some of the, those planes.

I'm going to show you later.

Let me move on to the, because it's going to be 430.

Um.

So here's the correlation.

So this is what I'm, I was saying before Daniel, that this is the, the piece, right?

Are these things, right?

The response is divided by the total number of questions.

Right.

So either I can talk about Rs or I can talk about P's.

And then correlation is defined as something that looks exactly like you would define it for distribution, but it's here defined for a sample.

Right.

So basically what I'm saying is I want to continue to write things as being individual products of performance.

And so what I really want to have inside this parentheses, right, is the number of times that they both said that it was a and it was a.

Well, that's this number here.

But I'm not going to put that number right.

What I'm going to do is I'm going to stubbornly stick with individual performance, and then I'm going to put the difference between that number that I need and what I, and this thing that I'm stuck with putting in, which is the product.

And so these are the four postulates for pair correlated binary classifiers.

You only need two of them.

You need one for the A label and you need one for the B label.

And it acts exactly like you would expect a correlation to do.

Right.

When, when the, when correlation is positive, it's going to increase the number of responses where they agree.

And when it's negative, right, it's going to increase when they disagree.

It has the right sign basically.

Correlation here is plus contributing plus.

And in these two equations where they disagree is continuing with minus.

Yes.

Something like a finite procedural recasting and reimagining of the true positive, true negative, false positive, false negative.

Those are four actual values.

And so again, within the finiteness of digitized sensing, especially, you can take purely non-parametric, purely algorithmic, purely arithmetical, numerical approaches, rather than always passing everything through the kind of statistical meat grinder.

I'm done.

I'm done, Daniel.

I'm done.

I'm done.

You said, you said exactly.

Right.

This is, this is the thing.

Right.

And so I'm not going to go back into Tom Mitchell and Platonius.

One of the students came up with an independent solution, which is wrong.

And, and I've been slammed by reviewers because they say they, they did it right and I'm doing it wrong.

And it turns out to be the other way around, but I'm not going to go into that.

That's my personal beef.

But let me then show you what happens when you do, when you take these equations.

Right.

So, so what is the problem?

The problem that everybody's encountered with, with pair binary classifiers is that correlation is entangled with individual performance in every one of these equations.

Do you see that?

26A, 26B, 26C and 26D have correlation in every single one of them.

We cannot separate correlation from individual performance.

But if you use algebraic geometry, you can, and that's what I'm not going into.

You can separate these equations.

You get back the individual equations, 33A and 33B are exactly the same equations that I showed you before.

But then you get these new ones where you've decoupled correlation.

So 33D, 33E and 33F are the only ones that have correlation.

And it turns out that they're the same possible.

You get no information from them.

So, and in fact, then everything in binary classification is controlled by this number.

The number of times that you agreed in the B label minus the product of times that you said that it was B.

It turns out that you can prove that for binary classification, this is exactly the same as this other quantity.

And the way that I'm going to use error correlation is this one, this equation right here.

And the only thing I want you to note about this equation is that it's linear in the correlation.

So if you tell me what you think the grade is, and I observe how that you voted and how you differed in your voting, I can tell you what your error correlation is.

So I'm racing ahead and I'm just want to show you some evaluations.

Okay, so.

So.

So.

Let me see what the.

So this equation is the solution for the prevalence.

Of A's.

When they do a classifiers are making error independent mistakes.

It comes from this quadratic polynomial.

That you get from solving that very complicated system that I show at the end.

So now I need eight of them.

I need three of them.

Right.

So I need eight equations.

So now I have I, J and K.

And you see sort of a descending, a symmetric pattern, right?

This ladder goes down one way and it goes up the other way.

These are the B's and these are the A's.

And when you solve that equation, you get a quadratic.

And it's that quadratic that I'm excited to show you.

Right.

This.

Do you get a quadratic equation which has these complicated factors that depend only on the statistics that you have observed?

There's no free parameters.

So this is not Bayesian estimation in any way.

Right.

You have no freedom whatsoever.

So then when I do synthetic data where I I've made it to have 19, 20 of one of the variables, it predicted exactly.

But let me show you some experimental results.

And the one that I want to show you is where it fails.

That independent solution is going to fail if they if they if the classifiers are too correlated.

It's going to give you an imaginary solution.

Can I make a comment on that?

Yes.

Sorry.

Go ahead.

That is an imaginary solution is that the parabola doesn't touch the axis.

parabola doesn't touch the axis and gosh I look at this picture and I go I can draw this circuit I can make this circuit physical circuit right so that I could because right I could make a geometric construction that draws uh uh that draws this parabola and then I can check to see whether electrically it touches that axis and this is a warning light if you don't touch that axis your classifiers are correlated too much right the error independent solution has failed so this is something that no probabilistic method can do right no probabilistic method that assumes error independence can show that the error independence assumption is violated that's the main problem with distributions and statistics right if you make assumptions you cannot prove right because you're you're you're tuning your parameters to satisfy your assumptions but when you're doing things algebraically like I'm doing here there is no freedom you can be in a state where they they and and actually I can show because I can have a complete representation where I include all the correlations I can prove that when I do these computations if the classifiers ever have a square root that's unresolved this is the independent algebraic evaluator I can prove that that means that that set of classifiers is actually correlated it has non-zero correlation and below you're seeing

that that that's in fact the case right you're seeing that one of the the the correlations is almost
six percent and I've been able to detect with the independent solution that in fact there is non-zero
correlation because I have an unresolved square root that doesn't make any sense right these are integer
numbers what does it mean to say that that you're you got a percentage that is some square root
right and this cannot be done by any infinite statistical right this is the distinction I want
to make there's finite statistics which is about samples and there's infinite statistics which is an
infinite number of samples right and there is finite statistics where you where you accept that what's gold
what's gold is the sample is the sample is the sample is the sample is true everything that you derive from it is
is not possibly true versus the infinite statistical world
where the world the sample is always wrong right in statistics a given sample is always wrong
right the equality only occurs and the average of the samples
that's a completely different way of thinking yes and I say that the finite one is the one that I want
to choose in terms of safety and it's 4 30 exactly so I'm going to stop because you know I think I feel
like I've been talking too much and maybe people can chime in and push back and have interesting questions
yeah anyone in the live chat and then Jakob go for it
yeah it's um
I want to show you how I just want to say how crazy this is I'm giving you error correlations which are the only
logical consistent error correlations once I decided to do majority voting or not
and here I'm showing you the majority voting is kind of like a little bit too hysterical it thinks that
they're 25 percent correlated and in fact they're only at most one percent and and so my method right of
using the postulates gives a better self-consistent solution I haven't gone into that right but I go
what basically what I do is I take this number which is not at all an integer ratio right and I find the integer
ratio that's closest to it right and then I ask for that integer ratio what is the only
logically consistent error correlation and I get this these numbers yeah and that that reminds me of the
finite pi like 22 over 7 is a classic small approximator of pi and you can go into finite approximators
exactly and therefore get closer and closer incessantly keeping the strong guarantees right but at some
point it's useless you're you're you're you're doing that the width of a hydrogen atom as people like to
point out right your error in the circumference is going to be the width of a hydrogen atom who cares
right and so a finite heuristic is going to do it very well sorry Jacob I had to I had to interrupt
you there to just to point out right that that that the fact that I can even measure error correlation
and say that it's logically consistent is incredible I just have to say that
yeah and no worries I I'm still trying to wrap my head around the concept of disentangling the
semantics from the particular tool that you're using right um because I feel like it also kind of depends on
on your expectations as the person using these tools like going back to the going back to the example of
like a or b like if you if you take away um if you take away the particular content of those of those
questions and you only look at the statistics of the of the answers of correctness I would say statistics
of correctness statistics of correctness then perhaps you perhaps you are taking away the correlation from
the particular content of those questions yes yes are you taking away the correlation of for instance

the person who said those questions oh my god it's so good it's so good that you were saying right so so
so here's what I had to do at UMass when I was giving the the exams right so so UMass Amherst is a state school right so you get bubble sheet answers right uh exams at UMass Amherst right but I taught at Williams and at Swarthmore right I never gave bubble sheet exams at Swarthmore or Williams right so it's only right that when you have large classes that you have these things right and so at UMass right
the students would cheat so they told me you have to make five exams and dress the student is one color exam
and then you put you see people in a diamond pattern around around that student
so that everyone around the student has a different color exam and then you scramble the the the responses
so nobody can look over anybody's shoulder and copy the bubble right so what am I trying to do there why am I
trying to do that because I want to prevent the error correlation that occurs because you cheat it
and I'm trying to find out the error correlation between all the students that is due to a commonly
shared cognitive misunderstanding of Newtonian physics maybe they're still thinking right maybe I put a question that question that is testing their their belief in Aristotelian physics right and I want to detect
that that and so when I get a wrong answer in a question the purpose of the exam for me as a professor
is to find out what I should be talking about the next class right to address the mistakes that people are making
cognitively I want to design a test that blocks any error correlation except the cognitive one
and so that goes to what you're saying Jacob right you get to design what error correlation right
somebody's error correlation right maybe a gold for you right I as a teacher want to see cognitive error correlations because I want to help you and teach better and the when I talked about digitizing
you you brought up another thing Jacob which is that maybe in the original signal
the responses were not correlated but then when I digitize it I correlate it
I have no and and I think Daniel also talked about that right that when you change formats right
you're introducing spurious signals and all of them are in there and yes there is no guarantee right but
but by the way uh there is the advantage here that I can measure the error correlation
right so that you can see whether I did introduce right by digitizing the the logical consistency of your grading if I introduce correlations or not that existed in the original format does that make sense
yeah there's no free lunch you won't get anything for free here if you really think about it right it sounds magical but it isn't
it's okay on the free lunch theme kind of thinking about this in a fun way there's a lunch where you just show up and it's there for you
so that's kind of the dream scenario if you're one of those who eat lunch at that time because it's kind of like you've chosen
the axis to create a finite projection to you've the it's the right test the seating arrangement
there's just an ideal situation and then some of those may be actually perhaps even relaxed in practice
like I I don't know all the details but when you described that you're you taking an integer approximator
so there's ways of of in of integerizing and finitizing that strip semantics but then I thought about what if the

ship of fools what if none of them can be the captain

well then the ship's going to go extinct and then it doesn't matter what that's the imaginary parabola that doesn't make contact

it's like no there's no decision rule there's also no quadratic regression but this is just an imaginary solution

these people are talking about survival strategies that are imaginary

yes yes so so you get the other part about it which is what why it was good to put it in a philosophical uh journal

which is what if there is no answer key what if we are all deluded and we're just testing with a test our

alignment on a delusional belief i explained this to uh uh an ex-wife of mine who's a psychoanalyst

and she because i was explaining the concept of logical consistency versus logical soundness by the way

with logical soundness you need to know the truth table logical consistency you don't you're just checking

to see right without semantics that you're that you're uh cranking the logic machinery correctly

right if your premises are correct your conclusions are correct so that's what i mean here by logical

consistency of grading right i cannot tell you exactly what the grade is but i can tell you whether

you're being logically consistent with how you believe uh your your experts are behaving

right so that that is yes

that's the only thing that can be checked here right but it can be probably shown right

that that you are in a highly correlated state and therefore you should raise an alarm

and and so when you look at at uh what this could be it could be also you could think of it as a way

of selecting right if you have different models and you find the one that's most aligned with what your

beliefs are about what the truth is then you have instant compression right just select that model

right because you found the one that's most aligned with everybody right and so you don't need to keep

all of them but there's nothing to right to say that you're not hallucinating and that you're going to

be eaten right because you think that there is no tiger there but there is a tiger

right you could have false beliefs and be delusional right confident and alive not confident and soon to

disease exactly right and so there is nothing here that prevents that from happening but it would

allow you if you were initially in a good operating state to start detecting something that is malfunctioning

right so that you could prevent that from happening yeah i'll ask a question from the live chat

upcycle club writes i have a question how do you define the data sketch of the decisions

so that was in the uh that was in the uh the the statement about completeness right so whenever you

look at an ensemble uh there by the way there's there's a if you know data streaming right there's

data streaming algorithm for every statistic right sample statistics there's many of them

of course with a finite sample there's only a finite number but there's many of them right combinatorially

explosive so here it is right if i'm talking about three right i have to talk about two to the three

or eight different frequencies if i'm looking at only their alignment right but that's not the only

statistics i may want to right suppose that you're looking at classification of of uh dna pairs so you're

worrying so there's also sequence information so you may want to know how accurate you are on the point

you know on the on the given base pair but you may want to know what what what percentage of times

you were correct on both pairs so now you're keeping tab of another set of statistics which is not the
one here in equation 38 right because now you're comparing uh two different points on the dna sequence
right and so you could be keeping statistics about the observation 10 steps back or 10 steps forward
right so there's an infinite number of statistics that you could be doing so i've only shown one
small set of them and for every one of them you're going to be able to write this summation right it's
it's a part of uh if you will of the basis of probability theory that you have event space right and
that you can delineate event space completely right so you have completeness right there is no event that
you do not know about if you and by the way there is no theory about all the theories of the world
this is why you cannot do this with theoretical work right because we do have no theory that can
enunciate all the postulates of all the possible theories about the world this can only be done with
sample statistics right that's how i get away with this yeah to to kind of reflect that back i i first
off i again i appreciate this presentation and work i i encourage people to to look into this because
it's very interesting and to connect to the prior century of statistics and understand where exactly
this is positioned and and how deeply this statistical paradigm is baked into the safety discussion and
to this last point that you're creating many possible finite spaces um the error patterns
and covariances can be harder to fake than the mean you could just take one number oh my god daniel
oh my god okay oh my god i mean i mean come on it just yes exactly right right you can fake the signal
but you cannot fake the correlation it's so much harder right it's so much harder to do that
yeah if they're faking the mean and the variance and all the higher coordinates of motion then it's
indistinguishable exactly exactly right yes so this applies to the concept of enemy inside the gates
how would you detect that something inside of you is hallucinating right now and you shouldn't be
paying attention to it right this is would be a crucial thing for any right let's go back to the
original question right forget about what i said if it's not how i said it is it has to be something
like this how can an entity have intelligence have integrity of thought if it doesn't have a way of
detecting enemy inside the gates yes you have to defend against yourself
plugged in security yes right this is this is what i think is so interesting about this
right forget about right the number one enemy you have to worry about is you
your hallucination that is preventing you from seeing the tiger
that i would think that that would be the primary thing and like going back to the semantics i can
see how this can be used for binary classification of right of safe unsafe right tiger not tiger
but then because the mathematics is exactly the same how the a biological system could co-opt exactly the same
computational structure to just check correctness of very complicated opinions not just binary classification
how about jakub um a last thought and then we'll just each have a closing word here
yeah i'm trying to connect these thoughts to um i guess the the active inference formalism where like
where how many how many layers of uh of metacognition do you need to um to re to realize or
elicit this type of self-reflective self-reflective behavior on the different systems that you're
modeling to all of them right this would be used everywhere like crazy right you could use this over

and over and over and over and over and over right to check every single component
right it has no intelligence so therefore it can be used right anywhere that's what i'm saying that
this is the thing that i find fascinating about it because it has no intelligence and it stripped the
semantics you can insert the instrument anywhere to check statistics of correctness of your noisy
oracles but what that makes me think about is uh chris fields's recent course andre's not not
saying you watch this or anything like that but what he spoke to was that there's a classical screen
that can be digitized a physical screen and then there's two agents that have these semantic reference
frames that was done in a bayesian statistical setting and you have kind of um added discrete beads
to some first approximation inside some of these otherwise distributional considerations
like if the two thermometers can be from zero to ten that's how many paralyzed observations there are
you can have all kinds of distributions and flows and that type of modeling and continuous state spaces
or there may be a whole variety of activities available in the discrete cube including these things like
well one over and seven up and then that correlation again to make the point clear
the point clear it can be basically detected and and triangulated where discrete differential is occurring
in active inference we have the kind of continuous optimization machine learning but this is more
like pinpoint like this bolt is failing rather than like there's a 0.8 that one of the bolts is loose
no it's this this is failing yes it's it's definitely trying to identify this is failing don't pay
attention to right maybe take it out of the ensemble what are the next steps for your work
so the so kind of going through the binary case that has made me realize it was very hard to figure
out the binary case i've been working on it since 2010 but now that i've finally gone through the other
end that i made and i figured out how to make it simple i'm starting to be able to write down what
the solutions have to be for the three label case so the r equals three test and so what i've not gone
into is that it goes back to that issue of these two numbers being the same 34 and 35 there's no more
information in the binary case but in the in the three label case you would find that these three
things are not the same right so strangely enough when you have more responses there's more information
because there's more chance of disagreement and disagreement is the most informative thing that
you can have for an ensemble agreement is useless agreement is a tiny portion of these equations right
look at these equations these equations there are four of them right well not this one but in the in
the eighth one you can see it yeah the maximally informative coin flip is 50 50. that's the statistical
way to say this right but here right there are eight equations and only two of them are when they
completely agree and this is what people like to talk about plutonius and tom mitchell talk about the
agreement equations where they just wanted to talk about 41 plus 48 at it and i'm like why why not
look at the full set of information that's available when you consider 41 42 43 40 you know what i'm saying
each one of these is true it's much more complicated when you have correlation then you have pairwise
correlation and then you have three-way correlation but my point is right that these things are are are
quite complicated and but but but they have a very predictable structure this makes me think about
it being a meta observation yes it's a second level it's an actual
calculated algorithm it's an actual fact as much as the thermometer's reading is a fact

the thermometer that measures the difference between them is also a fact

that's right and then you have to decide well i know that i'm in this domain

and you know this is temperature and temperature is not going to change very quickly because it's a car

engine right so it can't actually fluctuate you know what i'm saying that kind of thing so if you see

something that's fluctuating too much you know that it's wrong right that's domain knowledge right

being applied to then take a measurement right and then interpreting that measurement as being a

a safety issue or not right again the computer is needed right the thermometer is providing the

reading but the computer is needed to understand why the other analogy i like to make is like a murder

mystery a crime has been committed with this method you can interview all the suspects and find the

illogical consistency the suspect right because they have a story that doesn't compare to anybody else

but you cannot find why and how because there's no semantics of why people commit murders or how they commit them

right so this helps you solve the crime who committed the crime you know this this algorithm is malfunctioning

but it cannot help you figure out why

or how right that you need a higher cognition right

thank you again for joining thank you for inviting me in this wonderful discussion

indeed perhaps a 67.2 in a future time yes thank you thank you guys thank you

thank you

Thank you.